

Md. Mahmudul Hasan

PhD Student (First Year), Division of Systems Engineering, Boston University

Assistant Professor (On-leave), Department of Electrical & Electronic Engineering (EEE)

Rajshahi University of Engineering & Technology (RUET), Rajshahi- 6204, Bangladesh

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Research Interests

- Multi-objective metaheuristic optimizations for controller tuning and energy management in renewable-integrated power systems.
- Disturbance observer-based control system design for renewable energy applications.
- Virtual inertia emulation and frequency stabilization techniques for converter-dominated microgrids.

Education

Boston University

August 2026 – Present

Ph.D. in Systems Engineering

- **Coursework (Continuing):** Dynamic System Theory, Optimization Theory II, Probability with Statistical Applications

Rajshahi University of Engineering & Technology

April 2020 – December 2024

M.Sc. in Electrical & Electronic Engineering

- **CGPA:** 3.92/4.0
- **Thesis:** Development of a Hybrid Metaheuristic Optimization Based Control Platform for Three Phase Microgrid System
- **Coursework:** Engineering Analysis, Modern Control Theory, Multivariable Control Systems, Renewable Energy, Energy Conversion, Digital Circuit Design

Rajshahi University of Engineering & Technology

January 2015 – August 2019

B.Sc. in Electrical & Electronic Engineering

- **CGPA:** 3.89/4.0 (1st class 2nd among 120 students)
- **Thesis:** Artificial Neural Network Based Sensorless Field Oriented Control of Three phase Induction Motor
- **Major:** Power System

Academic Positions

Assistant Professor

Rajshahi- 6204, Bangladesh

Department of EEE, Rajshahi University of Engineering & Technology

January 2024 – Present

- **Taught Courses:** Measurement and Instrumentation, Computational Methods in Engineering, Renewable Energy, Power System Operation and Control
- **No. of Undergraduate Thesis Supervised:** 5
- **Additional Role:** Member Secretary, OBE Based Undergraduate Curriculum Preparation Committee

Lecturer

Rajshahi- 6204, Bangladesh

Department of EEE, Rajshahi University of Engineering & Technology

February 2021 – December 2023

- **Taught Courses:** Measurement and Instrumentation, Power Plant Engineering and Economy, Power System II, Power System Operation and Control
- **No. of Undergraduate Thesis Supervised:** 3
- **Additional Role:** Member, Undergraduate Curriculum Modification Committee

Lecturer

Dhaka- 1216, Bangladesh

Department of EEE, Prime University

October 2019 – January 2021

- **Taught Courses:** Electronics I, Electromagnetic Fields, Digital Electronics and Pulse Techniques, Electrical Machine, Renewable Energy, Power System Operation and Control

Publications

Journal Articles

- [1] **M. M. Hasan**, M. S. Rana, T. S. Joyeeta, and F. Tabassum, "Virtual inertia emulation in islanded microgrids using a time-series disturbance observer for frequency stabilization," *International Journal of Electrical Power Energy Systems*, vol. 176, p. 111673, 2026. DOI: <https://doi.org/10.1016/j.ijepes.2026.111673> .
- [2] R. Hasan, **M. M. Hasan**, M. S. Rana, F. Tabassum, and M. N. B. Reza, "Disturbance observer-aided SSA-TID control for frequency stabilization in hybrid renewable power systems," *IET Renewable Power Generation*, vol. 19, no. 1, 2025. DOI: doi.org/10.1049/rpg2.70137 .
- [3] F. Tabassum, M. Rana, T. S. Joyeeta, and **M. M. Hasan**, "Optimal defense against security threat on stability of grid-integrated synchronous generator during fdi attack," *e-Prime - Advances in Electrical Engineering, Electronics and Energy*, vol. 13, p. 101097, 2025. DOI: [10.1016/j.prime.2025.101097](https://doi.org/10.1016/j.prime.2025.101097) .
- [4] M. N.-U.-Rahman, **M. M. Hasan**, A. M. Raka, *et al.*, "An asymmetric fuzzy-based self-tuned pso-optimized mppt controller for grid-connected solar photovoltaic system," *Energy Conversion and Management: X*, vol. 26, p. 100902, 2025. DOI: [10.1016/j.ecmx.2025.100902](https://doi.org/10.1016/j.ecmx.2025.100902) .
- [5] **M. M. Hasan**, M. S. Rana, F. Tabassum, H. R. Pota, and M. H. K. Roni, "Optimizing the initial weights of a PID neural network controller for voltage stabilization of microgrids using a PEO-GA algorithm," *Applied Soft Computing*, vol. 147, p. 110771, 2023. DOI: [10.1016/j.asoc.2023.110771](https://doi.org/10.1016/j.asoc.2023.110771) .
- [6] M. H. K. Roni, M. S. Rana, H. R. Pota, **M. M. Hasan**, and M. S. Hussain, "Recent trends in bio-inspired meta-heuristic optimization techniques in control applications for electrical systems: A review," *International Journal of Dynamics and Control*, vol. 10, no. 3, pp. 999–1011, Jan. 2022. DOI: [10.1007/s40435-021-00892-3](https://doi.org/10.1007/s40435-021-00892-3) .
- [7] M. R. Islam, J. Hasan, **M. M. Hasan**, M. N. Huda, M. A. H. Sadi, and A. AbuHussein, "Performance improvement of dfig-based wind farms using narma-l2 controlled bridge-type flux coupling non-superconducting fault current limiter," *IET Generation, Transmission and Distribution*, vol. 14, no. 26, pp. 6580–6593, 2021. DOI: [10.1049/iet-gtd.2019.1917](https://doi.org/10.1049/iet-gtd.2019.1917) .

Journal Articles Under Review

- [1] **M. M. Hasan**, M. M. Hasan, M. S. Rana, F. Tabassum, and T. S. Joyeeta, "Multivariable power control of grid-forming inverters using a PID neural network," Manuscript under review at *IET Renewable Power Generation*, 2025.

Selected Conference Papers

- [1] M. N. M. Rahi, **M. M. Hasan**, M. Y.-Y. U. Haque, A. I. Abeg, and T. S. Turna, "Voltage control of grid-forming inverter in islanded microgrids using PSO-optimized sliding mode controller," in *Proc. Int. Conf. Electr., Comput. Commun. Eng. (ECCE)*, 2025, pp. 1–6. DOI: [10.1109/ECCE64574.2025.11013018](https://doi.org/10.1109/ECCE64574.2025.11013018) .
- [2] M. A. Hossain, **M. M. Hasan**, F. Tabassum, and M. S. Rana, "Salp swarm algorithm based integral LQG control of two area nonlinear power system with renewable energy penetration," in *Proc. Int. Conf. Comput. Inf. Technol. (ICCIT)*, 2024, pp. 657–662. DOI: [10.1109/ICCIT64611.2024.11022531](https://doi.org/10.1109/ICCIT64611.2024.11022531) .
- [3] **M. M. Hasan**, M. S. Rahman, M. S. Rana, and F. Tabassum, "Optimal fractional order control of virtual inertia emulator for a microgrid with high renewable energy penetration," in *Proc. Int. Conf. Dev. Renew. Energy Technol. (ICDRET)*, 2024, pp. 1–6. DOI: [10.1109/ICDRET60388.2024.10504017](https://doi.org/10.1109/ICDRET60388.2024.10504017) .
- [4] M. A. Rafi, **M. M. Hasan**, M. Y.-Y. U. Haque, Z. Ferdous, M. S. Rana, and M. F. Hossain, "Performance augmentation of an arc suppression device using particle swarm optimization based proportional resonant controller," in *Proc. Int. Conf. Electr. Comput. Eng. (ICECE)*, 2022, pp. 445–448. DOI: [10.1109/ICECE57408.2022.10089057](https://doi.org/10.1109/ICECE57408.2022.10089057) .
- [5] **M. M. Hasan**, M. S. Hussain, M. S. Rana, and M. H. K. Roni, "Population extremal optimization based 2-DOF control strategy for field oriented control of induction motor," in *Proc. Int. Conf. Electr. Electron. Eng. (ICEEE)*, 2021, pp. 117–120. DOI: [10.1109/ICEEE54059.2021.9718799](https://doi.org/10.1109/ICEEE54059.2021.9718799) .

- [6] **M. M. Hasan** and M. S. Rana, “Modified population extremal optimization based PID neural network for islanded microgrid system,” in *Proc. IEEE Int. Conf. Comput., Power Commun. Technol. (GUCON)*, 2021, pp. 1–6. DOI: [10.1109/GUCON50781.2021.9573922](https://doi.org/10.1109/GUCON50781.2021.9573922) [↗](#).
- [7] **M. M. Hasan**, “Artificial neural network based speed estimator for sensorless field oriented control of three phase induction motor,” in *Proc. Int. Conf. Electr., Comput. Telecommun. Eng. (ICECTE)*, 2019, pp. 57–60. DOI: [10.1109/ICECTE48615.2019.9303559](https://doi.org/10.1109/ICECTE48615.2019.9303559) [↗](#).

Supervised Undergraduate Thesis

1. *A 2-DOF Optimal Active Disturbance Rejection Framework for Load Frequency Control*, M H M Tawhidul Islam Fahim and Fayeze Naqib Efat, Department of EEE, RUET, 2025.
2. *Multi-Objective Optimization Based Energy Management System for Grid Connected Microgrid*, Susmita Paul, Department of EEE, RUET, 2025.
3. *Optimal Fractional Order Control of Virtual Inertia Emulator for a Microgrid with High Renewable Energy Penetration*, Md. Sohanur Rahman, Department of EEE, RUET, 2024.
4. *Salp Swarm Algorithm Tuned Fractional Order Controller for Load Frequency Control of Hybrid Renewable Power system*, Rakibul Hasan, Department of EEE, RUET, 2024.
5. *Optimal Active Disturbance Rejection Control of Power System Stabilizer*, Maliha Anam, Department of EEE, RUET, 2024.
6. *Robust Optimal Dynamic Voltage Restoration in a Compensated Resonant Grounded Distribution System*, Ramisa Fariha Mridha and Naimur Rahman, Department of EEE, RUET, 2023.
7. *Performance Augmentation of an Arc Suppression Device Using Active Disturbance Rejection Control*, Ragib Rowshon, Department of EEE, RUET, 2023.
8. *Voltage Control of PV-fed Grid Forming Inverter Based Three-Phase Microgrid Using Optimal Sliding Mode Controller*, Munsiri Nahid Md Rahi, Department of EEE, RUET, 2023.

Review Activities

- Reviewer, Electric Power Systems Research, Elsevier
- Reviewer, ISA Transactions, Elsevier
- Reviewer, International Transactions on Electrical Energy System, Hindawi
- Reviewer at different IEEE international conferences

Technical Skills

Optimization: Multi-objective metaheuristic optimizations, Convex optimization

Control System Design: State-feedback, observer based control, H_∞ control, Adaptive and neuro-adaptive control, disturbance observer design for linear and nonlinear systems, sliding mode control with observation

Deep Learning: Designing and training ANN, RNN, LSTM, and GRU

Programming: MATLAB, Python (PyTorch, Pandas, NumPy, Matplotlib), C

Simulation Tools: MATLAB/Simulink, Proteus, PSpice

Hardware Platforms: Arduino, Sensor integration, PCB design

Productivity Tools: LaTeX, Microsoft Office

Industry Standard Testing: Experienced in testing of power transformers (up to 800 kVA), solar PV module, insulators, lightning arresters, transmission line cable, and LED streetlights according to IEC standards.

UGC Funded Project

Project Title: Adaptive Control of Islanded Microgrid System using Population Extremal Optimization Based PID Neural Network

Role: Project Director

Grant No.: DRE/7/RUET/528(35)/PRO/2021-22/14

Duration: 2021-2022

Grant Amount: BDT 2,50,000

Industrial Training

Ghorashal Power Station

Bangladesh Power Development Board

Ghorashal, Bangladesh

08 May 2018 – 23 May 2018

- **Description:** The training was conducted on power generation (steam turbine, gas turbine, combined cycle, and gas engine power plants), substation, switchgear, and power plant simulator.

Standardized Test Score

IELTS Academic: Overall Band 8.0 (Listening: 9.0, Reading: 9.0, Writing: 6.5, Speaking: 7.0)

Professional Services

- Publication Secretary, IEEE International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), 12-13 September 2024. Venue: Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh.
- Publication Committee Member, IEEE 3rd International Conference on Electrical & Electronic Engineering (ICEEE), 22-24 December 2021. Venue: Rajshahi University of Engineering & Technology (RUET), Rajshahi, Bangladesh.
- Promotional and Publication Sub-committee Member, 1st Reunion of EEE Alumni Association, RUET.
- Member, Electrical Testing Committee, Department of EEE, RUET.
- Invited Talk: “Energy Management and Virtual Inertia Emulation in Microgrids”, IEEE PES Day, IEEE Student Branch, Bangladesh University of Business and Technology (BUBT), 2024.

Awards and Scholarships

- EEE Association Award for Academic Excellence, 2016
- Education Board Scholarship for Excellence in Higher Secondary School Certificate Examination, 2014
- Education Board Scholarship for Excellence in Secondary School Certificate Examination, 2012

References

Prof. Dr. S. M. Abdur Razzak
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Vice-chancellor, RUET
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Relation: Undergraduate supervisor

Prof. Dr. Md Sohel Rana
Professor, Department of EEE, RUET
Email: sohel@eee.ruet.ac.bd
Relation: Postgraduate supervisor